

The Analytics Edge

Summer 2026

Course Description

The availability of data has changed the way organizations are thinking about themselves and the way they interact with the world. Data is helping improve the profit of businesses, the quality of life of individuals, the performance of sports teams and in the ways we network with each other. In this course you will learn through cases on how to use analytics to give you an edge. The course will expose students to real world examples of how analytics has and is being used. Through these examples you will learn the fundamentals of and how to use tools of predictive and prescriptive analytics such as linear regression, logistic regression, multinomial and mixed logit, classification and regression trees, random forests, clustering and optimization in practice. We will use the statistical software R (for predictive analytics) and the numerical computing software Julia with optimization packages (for prescriptive analytics) for these tasks. AI tools can assist you with the implementation of models in these softwares. Our focus will be primarily on models that are more interpretable.

Learning Objectives

1. Identify the link between data and models to help make better decisions that add value to individuals, companies and institutions.
2. Describe data effectively, predict future outcomes and prescribe decisions using the tools of analytics.

Measurable Outcomes

1. Develop a mathematical model from a given dataset.
2. Solve and analyze the results of the mathematical model using the tools of analytics such as linear regression, logistic regression, multinomial logit, mixed logit, classification and regression trees, random forests, clustering and optimization.
3. Demonstrate an understanding of the fundamental mathematical ideas in these models.

4. Compare and contrast decisions from mathematical models and simple heuristics and/or expert opinions.

Books

1. Textbook: The Analytics Edge by Dimitris Bertsimas, Allison K. O'Hair and William R. Pulleyblank Dynamic Ideas, Belmont, Massachusetts, 2016. Notes and supplementary material will be provided each week.
2. The students can also refer to the online course on OCW by MIT - The Analytics Edge.
3. A related book for statistical learning is: An Introduction to Statistical Learning with Applications in R by Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, Springer, 2014.

Software

1. We will make use of the free statistical software R for the course.
Download at <http://www.r-project.org/>
2. We will use it in conjunction with RStudio.
Download at <https://posit.co/download>.
3. There are now AI assistants in RStudio to help with these codes:
Posit Assistant (runs with Claude) has a free trial version and a paid version.
<https://docs.posit.co/ide/user/ide/guide/tools/posit-ai.html>
Github Copilot is a free AI assistant.
<https://docs.posit.co/ide/user/ide/guide/tools/copilot.html>
4. We will also make use of the numerical computing software Julia for the course.
Download at <https://julialang.org/>.

AI Tool Policy

The landscape of how these data analytics tasks can be run has changed in fundamental ways over the past few years. Today users can prompt and obtain codes without needing to know all the details of the syntax. In this course, you are free to use AI tools to help with the codes - most of the data used in the code is open source. I will not be testing you in the exam on the development of codes. However always remember to analyze the output of the models returned by any such tool to see if it is doing the desired task.

Prerequisites

Statistics and optimization

Course Assessment

Assignments	10%
Mid-Term Test (Week 6)	30%
Competition (week to be confirmed)	30%
Final Test (Week 14)	30%

Instructors

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For consultation, just email me and set up a time.

Timetable

Cohort sessions - Think Thank 2 and 3

CS01 - Tuesday 13:30-15:30 and Thursday 16:00-18:00

CS02 - Monday 13:00-15:00 and Wednesday 11:00-13:00

CS03 - Tuesday 16:00-18:00 and Thursday 13:30-15:30

Recitation - Lecture Theatre 2

Thursday 18:00-19:00

Exams

26 June (Friday) (Midterm)

19 Aug (Wednesday) (Final)

Topics to be covered (Tentative)

Before Midterm

Topic 0 Old Faithful

Topic 1 Method: Linear Regression
Predicting the quality and prices of wine (**Wine analytics**)
Moneyball (**Sports analytics**)

Topic 2 Method: Logistic Regression
Space shuttle failures (**Space analytics**),
Framingham heart study (**Healthcare analytics**)

Topic 3 Method: Discrete Choice
Oscars (**Movie analytics**)
Safety features in cars (**Transportation analytics**)

Topic 4 Methods: Sparsity and Model Selection
Baseball (**Sports analytics**)
Cross-country growth regressions (**Economic analytics**)

Topic 5 Methods: Discrete Optimization (**Capstone allocation**)

After Midterm Classification and Regression Trees (CART), Random Forests
Clustering
Matrix Factorization
Censored Data Analysis
Optimization